Cool Pavements for Sustainable Urban Development

Sushobhan Sen[a,1], **Jeffery Roesler**[a], **Benjamin L. Ruddell**[b], **and Ariane Middel**[c]

[a]University of Illinois at Urbana-Champaign, Urbana, IL 61820, USA; [b]Northern Arizona University, Flagstaff, AZ 86011, USA ; [c]Arizona State University, Tempe. AZ 85287, USA

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Around the world, urban development and densification leads to the Urban Heat Islands (UHI) effect, in which cities are warmer than adjoining rural areas. Cool pavements have been recommended as a mitigating strategy for the UHI effect. However, the spatial extent over which cool pavements need to be applied to achieve widespread mitigation has received little attention. A previously developed urban microclimatic model for the Power Ranch community in suburban Phoenix, Arizona, was used to investigate this question. The microclimatic model is used to investigate the effects of urban densification for the meteorological conditions at 5:00 PM on August 13, 2015. In the modeled scenario, the heights of the buildings were increased from 5 m to 10 m, a large, central park was redeveloped as a parking lot, and a reflective pavement was implemented in the parking lot. Both localized and downwind air temperature effects at 2m of this further densification were quantified in the modelling effort. For the lower building height, using typical concrete to redevelop the park as a parking lot increased the 2 m air temperature directly over and downstream by about 0.20 ℃. When a reflective concrete parking lot was used instead, the 2m air temperature decreased by 0.20 ℃ over and downstream. At 10 m building heights, the reflective parking lot decreased the 2 m air temperature by 0.20 ℃, however, its effect was more localized with less benefit for downstream areas. Thus, urban form with taller buildings affects the airflow, which requires a more distributed application of reflective surfaces to mitigate UHI.

Urban heat island| parking lots| computational fluid dynamics| reflective pavements

1. INTRODUCTION

The global population that lives in urban areas has been increasingly steadily over the last century, and in 2008, more than half the world lived in urban areas for the first time (Seto & Shepherd, 2009). Increasing urbanization has led to several challenges, ranging from public health (Kovats & Akhtar, 2008; Tao, Li, Crittenden, Lu, & Sun, 2016; Civerolo, et al., 2007) to resource consumption (Madlener & Sunak, 2011; Wang, Zeng, & Wu, 2016). One of these challenges is the development of the Urban Heat Island (UHI) effect, which is manifested in the form of urban areas having higher air temperatures than adjacent rural areas. The UHI effect has been observed in hundreds of cities around the world (Tran, Uchihama, Ochi, & Yasuoka, 2006; Peng, et al., 2011) and has been the subject of decades of study (Arnfield, 2003).

There are several causes for the UHI effect, as outlined by (Kleerekoper, Van Esch, & Salcedo, 2012). Two of these are the substitution of natural surfaces with impervious ones that have a lower albedo, and a reduction in wind speed and turbulent heat transport as a result of taller buildings. The relationship between land cover and UHI has been extensively studied through remote sensing (Chen, Zhao, Li, & Yin, 2006; Buyantuyev & Wu, 2010; He, Liu, Zhuang, Zhang, & Liu, 2007). Impervious surfaces, which include roads, roofs, and buildings, absorb more heat than natural surfaces because of their lower albedo and store that heat for a longer period, which leads to higher air temperatures around the surfaces and hence the UHI effect. Other studies have shown the relationship between urban form and the UHI effect (Stone Jr. & Rodgers, 2001; Zhou, Rybski, & Kropp, 2017; Rajagopalan, Lim, & Jamei, 2014). Taller buildings create deeper urban canyons, in which low wind speed leads to the recirculation of warm air that is not readily advected away by cooler air. This further increases the air temperature in cities and magnifies the UHI. One recent study (Sen & Roesler, 2019) demonstrated that the contribution of low-albedo pavements and urban canyons to UHI are coupled, creating a vicious cycle of higher air temperatures in cities.

One of the methods suggested in literature to UHI is the use of cool surfaces, such as cool roofs and cool pavements, which have a higher albedo that typical construction materials. A review of several papers on these topics were conducted by (Qin, 2015; Santamouris , 2013; Akbari & Matthews, 2012). A previous study by the authors (Sen, Roesler, Ruddell, & Middel, 2019) investigated the effectiveness of applying cool roofs, walls, and pavements to mitigate UHI in the Power Ranch suburban community in Phoenix, Arizona. They found that cool surfaces could decrease the air temperatures, however, the area which the reduction could be realized depended on the heterogenous land cover and urban form of the community. In other words, cool surfaces could in general mitigate the UHI but its impact can be somewhat localized in the community.

2. RESEARCH SIGNIFICANCE

The present study extends previous work by the authors to study the impact of land cover and building height on the effectiveness of cool pavements. While several studies in literature have commented on the benefits of using cool pavements to mitigate UHI, less attention has been given to the extent to which those benefits are spread over an area with heterogeneous land cover. For example, (Santamouris, et al., 2012) examined the UHI mitigating potential of cool pavements in a park by the sea in Athens, Greece and found that, close to the coast, there was little benefit (in terms of reduction in air temperature) from using cool pavements, but that benefit improved further inland.

In this study, two potential consequences of urbanization will be studied: an increase in the height of buildings, and

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[1]To whom correspondence should be addressed. E-mail: sen6@illinois.edu



Fig. 1. (a) Location of Gilbert in the Phoenix Metropolitan Area; (b) The Power Ranch community, with the central park marked in the center

the redevelopment of a park into a parking lot. The UHI impact reflective pavements for these two scenarios will be investigated using tools developed in (Sen, Roesler, Ruddell, & Middel, 2019) in the Power Ranch community in the Gilbert suburb of Phoenix, Arizona. This study will provide guidance to engineers and urban planners on effectively implementing cool pavements to mitigate UHI.

3. METHODOLOGY

A. Study Area. Power Ranch is a community in Gilbert, Arizona, which is in the southeastern corner of the Phoenix Metropolitan Area, as shown in Figure 1(a). Phoenix has been the subject of several studies related to UHI (Chow, Brennan, & Brazel, 2012) because of its growing population, hot and dry climate, and increasing UHI. A map of the Power Ranch community is shown in Figure 1(b). The area is mostly composed of dense single-family residential units of 5 m height, with irrigated grass lawns. Several local roads and two collector streets are present in the study area.

The study area also consists of a central park, as marked in Figure 1(b), with a man-made lake. The park serves as a community center for the residents of the area and is adjacent to an elementary school. All the existing roads are made of aged asphalt with an estimated albedo of 0.22.

B. Model. A Computational Fluid Dynamics (CFD)-based model for Power Ranch was developed in a previous study by the authors to estimate the 2 m air temperature (Sen, Roesler, Ruddell, & Middel, 2019). The model discretized the study area and computationally solved the complete Reynolds-Averaged Navier-Stokes (RANS) equations as well as the Reynolds-Averaged temperature equation, as shown in Eqs. (1)-(3), where U_i the wind velocity component in the direction of x_i, P is the kinematic pressure, T and T_{ref} are the air temperature and reference temperature respectively, ν and ν_t are the kinematic viscosity and turbulent kinematic viscosity respectively, α and α_t are the thermal diffusivity and turbulent thermal diffusivity respectively, g_j is the acceleration due to gravity in the direction of x_i, and β is the coefficient of thermal expansion of air.

$$\frac{\partial U_i}{\partial x_i} = 0 \qquad [1]$$

$$U_j \frac{\partial U_i}{\partial x_j} = -\frac{\partial P}{\partial x_j} + (\nu + \nu_t) \frac{\partial^2 U_i}{\partial x_i \partial x_j} - \beta (T - T_{ref}) g_j \qquad [2]$$

$$U_j \frac{\partial T}{\partial x_j} = (\alpha + \alpha_t) \frac{\partial^2 T_i}{\partial x_i^2} \qquad [3]$$

In addition, the surface temperature of the pavements, which is a function of the thermal diffusivity and albedo of the surface material, was calculated using a 1D heat transfer model called ILLITHERM. The pavement surface temperature provided a boundary condition to the CFD model. Using this model, the 2 m air temperature for the meteorological conditions at 5:00 PM on August 13, 2015 could be estimated and validated using field data from a mobile weather platform. At this hour, the far-field wind speed was 0.45 m/s from the east-southeast (ESE) direction, and far-field air temperature was 42.22°C. The same CFD model and meteorological conditions from (Sen, Roesler, Ruddell, & Middel, 2019), including the surface temperature of roofs, walls, and grass, were used in the present study for the cases described next.

C. Cases. A 2x3 factorial analysis was developed to study three scenarios for the area:

1. Building heights are 10 m instead of the existing 5 m.
2. The central park is redeveloped into a large parking lot with typical concrete of albedo 0.30.
3. The parking lot in scenario 2 is cast with reflective concrete of albedo 0.50.

The six cases investigated in the factorial analysis and their associated nomenclature are shown in Table 1 and presented below.

1. **Existing-5:** This case represents existing conditions, in which the central park is covered in grass and buildings are 5 m high. This case was taken from the validation results of the previous study (Sen, Roesler, Ruddell, & Middel, 2019).
2. **Existing-10:** This case was similar to the Existing-5 case, except that the height of the buildings was doubled to 10 m.

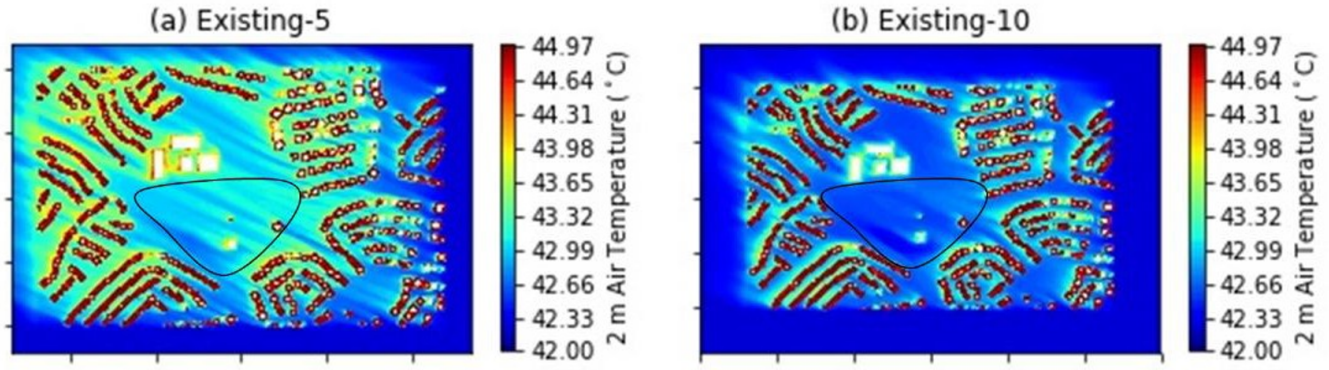


Fig. 2. 2 m air temperature (in ℃) in (a) Existing-5 and (b) Existing-10 cases, with the area above the central park marked

Table 1. Microscale UHI modeling cases investigated in the 2 x 3 factorial analysis

	Existing Grass Park	Typical Concrete Parking Lot (Albedo = 0.30)	Reflective Concrete Parking Lot (Albedo = 0.50)
Building Height = 5 m	Existing-5	Typical-5	Reflective-5
Building Height = 10 m	Existing-10	Typical-10	Reflective-10

3. **Typical-5:** This case was similar to the model from the Existing-5 case, but the central park was turned into a parking lot paved with typical concrete of albedo 0.30. All buildings were still 5 m high.

4. **Typical-10:** This was similar to the Typical-5 case, except all buildings were 10 m high.

5. **Reflective-5:** This was similar to the Typical-5 case, except that the parking lot was case with more reflective concrete of albedo 0.50.

6. **Reflective-10:** This was similar to the Reflecitve-5 case, except that the buildings were 10 m high.

Using these six scenarios, the effect of using reflective concrete as a cool pavement strategy in Power Ranch for mitigating UHI was investigated by first calculating the surface temperature of the parking lot pavements with the ILLITHERM model, and then the 2 m air temperature with the CFD model. The 2 m air temperatures were then compared to each other from each model run.

4. RESULTS

A. Existing Land Cover. The 2 m air temperature in the Existing-5 and Existing-10 cases are shown in Figure 2(a) and (b), respectively.

In the Existing-5 case, wind blowing from ESE at 42.22℃ was heated up by buildings and pavements, and this warm air was then advected downstream over the park. The average wind speed in the domain was 0.41 m/s and the air temperature over the park was about 43℃. However, in the Existing-10 case, the air temperature over the park was cooler, with an average air temperature of about 42.50℃ and an average wind speed of 0.37 m/s. Furthermore, downstream of the central park, the higher temperatures were more widespread in the Existing-5 case as compared to the Existing-10 case.

B. Parking lots with 5 m high buildings. The 2 m air temperature departures for the Typical-5 and Reflective-5 cases are shown in Figure 3(a) and (b), respectively. Temperature departure is the difference in air temperature from a reference value. The reference case for the Typical-5 and Reflective-5 cases was the Existing-5 case.

The redevelopment of the central park as a parking lot led to an average increase in 2 m air temperature over the park of about 0.20℃, on top of the increase from the buildings upstream. However, using a reflective concrete parking lot led to an average decrease in 2 m air temperature by about 0.20℃, reversing the upstream heating and mitigating the UHI. Downstream of the central park, the Reflective-5 case also showed a more widespread decrease in 2 m air temperature, while the Typical-5 case showed a widespread increase. In addition to downstream effects, upstream areas showed an increase in temperature due to thermal diffusion, which is a characteristic of incompressible air flow. In the Typical-5 case, warmer air from over the parking lot diffused upstream, while in the Reflective-5 case, the cooler air over the parking lot inhibited the warmer air from being readily advected away. These non-linear effects show the complexity involved in predicting the UHI impact of urban development.

C. Parking lots with 10 m high buildings. The 2 m air temperature departures for the Typical-10 and Reflective-10 cases are shown in Figure 4(a) and (b), respectively, with the reference temperature being the Existing-10 case.

In this case, the parking lot made of typical concrete led to an average increase in air temperature of about 0.30℃ over the parking lot, whereas using reflective concrete decreased the air temperature by about 0.20℃. Similar to the Typical-5 case, the Typical-10 case also saw a more widespread increase in air temperature downstream of the central park. However, unlike the Reflective-5 case, the downstream cooling in the Reflective-10 case was much more limited in spatial extent. Like the previous parking lot cases, upstream areas also experienced higher temperatures because of diffusion and inhibited advection.

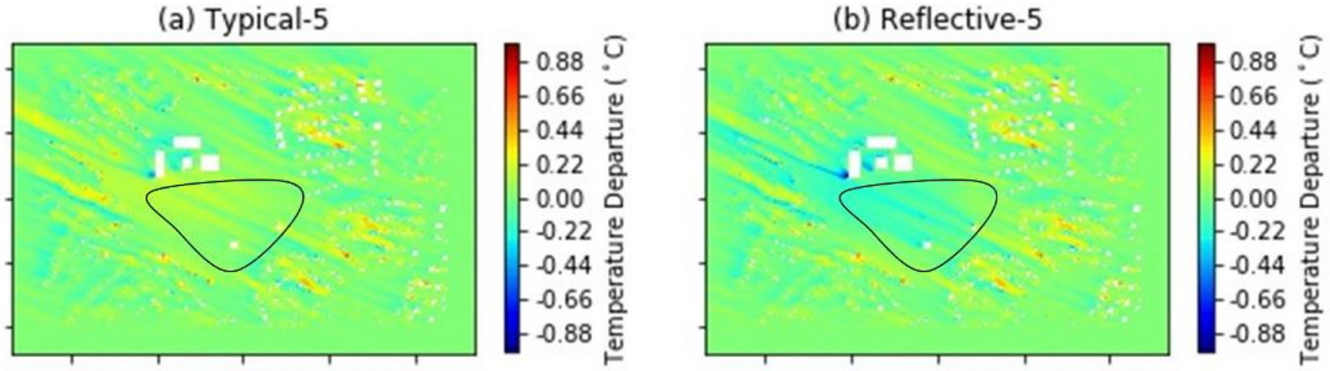


Fig. 3. 2 m air temperature departure (in ℃) in (a) Typical-5 and (b) Reflective-5 cases, with the area above the central park marked

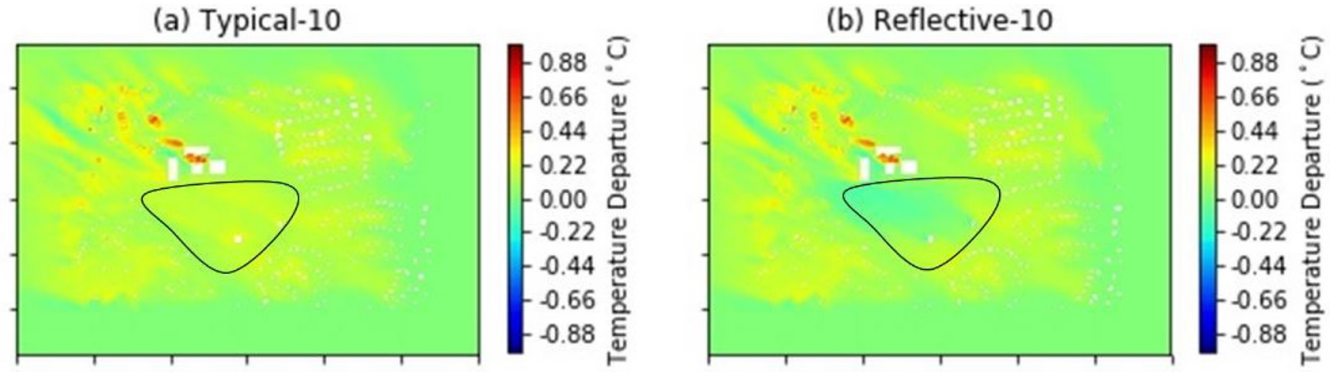


Fig. 4. 2 m air temperature departure (in ℃) in (a) Typical-10 and (b) Reflective-10 cases, with the area above the central park marked

5. DISCUSSION

From the results, it may be concluded, for existing buildings of height 5 m, that redeveloping the central park with a parking lot made of typical concrete increased the 2 m air temperature both immediately over the park as well as downstream of it by about 0.20℃, increasing the existing UHI effect in the community. However, applying reflective concrete as a cool pavement solution produced a significant reduction in the air temperature, which limits the impact of re-purposing the park to a parking lot. Note, this finding is not recommending replacing parking lots over the superior UHI benefits of vegetated surfaces (such as shading and latent heat) but a providing a cool pavement solution if redevelopment plan was underway to replace natural surfaces with man-made ones.

This beneficial effect of reflective pavements became more localized when the height of all the buildings was increased from 5 m to 10 m. Taller buildings increase the rate of turbulent dissipation, which decreases downstream wind speed. Thus, any heating or cooling effect from surfaces becomes more localized because of the slower advection of air. When the central park was redeveloped as a parking lot made of typical concrete, it increased the 2 m air temperature above the park and downstream of it by an average of about 0.30℃, as the slower air mass had more time to be warmed by the downstream pavements and buildings. Using a reflective concrete pavement for the parking lot decreased the air temperature by 0.20℃ immediately over the pavement, but there was only a limited decrease in air temperature downstream as the cooler air did not advect quickly and was re-heated by the downstream buildings and pavements. Furthermore, non-linear airflow effects led to an increase in air temperature upstream of the parking lot for both typical and reflective concrete pavement cases.

The most important implication of this study is that, for urban or suburban areas with low building height, a widespread reduction in UHI can be achieved by using reflective pavements over a relatively small but central area, with significant mitigation being experienced downstream of the prevailing wind direction. However, for taller buildings, the benefit of reflective pavements is more localized, and more distributed areas need to be made reflective to achieve the same level of UHI mitigation.

6. CONCLUSIONS

The UHI effect has been observed across the world and is caused partly by the replacement of vegetated surfaces with impervious ones, which absorb more heat and store it longer. This leads to a higher air temperature in urban areas as compared to adjacent rural areas. As the world continues to urbanize, the UHI effect is only expected to become more prevalent, posing a challenge to civil engineers and urban planners.

The Power Ranch suburban community in Gilbert, Arizona near Phoenix was investigated for the potential of mitigating UHI using reflective pavements while still allowing for increased urbanization in the form of a large parking lot being developed in place of an existing grass- covered park. Furthermore, the effect of doubling the height of all the buildings from the

existing 5 m was also analyzed using a 3D RANS-based CFD model to simulate the urban microclimate.

For the lower building height of 5 m, redeveloping the park with typical concrete of albedo 0.30 led to an average increase in 2 m air temperature of about 0.20℃ both over the park as well as downstream of it, further increasing the already existing UHI in those areas. However, using a reflective pavement with an albedo of 0.50 reversed this effect, decrease the average 2 m air temperature over and downstream of the park by 0.20℃. Thus, for areas with lower building height, a relatively small and centralized area of reflective pavements can lead to a widespread mitigation of the UHI effect both over the area and downstream of it.

When the height of the buildings was 10 m, the mitigating effect of the cool pavement was more localized because of an increase in turbulent dissipation and slower advection of cooler air. The parking lot made of typical pavements increased the air temperature by about 0.30℃ both over and downstream of the park. However, the parking lot made of reflective pavements again decreased the air temperature by 0.20℃ only over the park and a much smaller area downstream of it. This lower impact downstream was because the cooler air was advected slowly and was re- heated by the downstream buildings and pavements, thus localizing the benefit of the reflective parking lot. For widespread mitigation of UHI in areas with taller buildings, distributed areas of cool surfaces are necessary to avoid downstream re-heating of air.

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